S25-47-D-InterviewDen

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Session 2021 - 2025

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December, 2025

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Chapter 1

Introduction

Hiring in the IT industry has long been a challenging and time-consuming process. From manually reviewing countless resumes to conducting lengthy interviews, companies often struggle to find the right candidates efficiently. Human biases can sometimes influence decisions, making it harder to ensure fairness. On the other hand, job seekers frequently face uncertainty waiting for feedback and wondering how they performed during interviews. To address these issues, we developed and deployed InterviewDen, an AI-Driven Job Interview platform that successfully streamlines recruitment through automated resume screening, intelligent assessment generation, and adaptive AI interviews.

1.1 Existing Solutions

The recruitment industry has undergone substantial transformation with the integration of artificial intelligence technologies over the past decade. As organizations face increasing pressure to efficiently identify qualified candidates from large applicant pools while improving diversity and candidate experience, numerous AI-driven solutions have emerged to address different aspects of the hiring process. By understanding the state of existing solutions, we establish the foundation for how InterviewDen addresses persistent gaps in the market and delivers meaningful improvements to the recruitment ecosystem.

1.1.1 Resume Screening Solutions

- **Traditional ATS** (Workday, Taleo) offer efficient volume processing but rely on simplistic keyword matching with high false negative rates
- **AI-Enhanced Screeners** (Ideal, HireVue) provide more sophisticated matching but may perpetuate biases and lack transparency

- **Resume Optimizers** (Jobscan, VMock) offer improvement suggestions but focus on gaming systems rather than substantive enhancements

1.1.2 Technical Assessment Platforms

- **Coding Platforms** (HackerRank, Codility) provide standardized evaluation but often emphasize algorithmic puzzles over practical skills
- **Skills Assessments** (TestGorilla, Vervoe) cover diverse domains but typically offer generic tests not tailored to specific positions
- **Simulation Tools** (DevSkiller, Qualified) create job-relevant tasks but face standardization challenges

1.1.3 AI Interview Systems

- **Asynchronous Video** (HireVue, Spark Hire) enable convenient screening but lack natural conversation flow
- **Conversational AI** (Mya, Paradox) offer immediate engagement but have limited depth
- **Comprehensive Systems** (Interviewer.AI, Talview) provide multi-faceted evaluation but struggle with language diversity and transparency

1.1.4 End-to-End Platforms

- **Full-Service Platforms** (LinkedIn Talent, iCIMS) unify workflows but offer less specialized capabilities
- **AI-First Solutions** (Eightfold.ai, Phenom) feature advanced matching but require complex implementation

1.1.5 Key Market Gaps

- Fragmented stages rather than integrated experiences
- Limited actionable feedback for candidates
- Black-box evaluation lacking transparency
- Generic rather than customized assessments

- Imbalance between efficiency and candidate experience

- Inadequate bias detection and mitigation

1.1.6 InterviewDen's Differentiation

InterviewDen addresses these gaps through a unified AI framework, dual-portal approach, continuous feedback, transparent assessment criteria, dynamic customization, comprehensive bias detection, and integrated analytics that connect all recruitment stages for both efficiency and improved candidate experience.

Table 1.1: Competitor Analysis: AI Hiring for Tech-Based Industries

Feature/Platform	**InterviewDen (Tech-Focused)**	**HireVue**	**Micro1**	**Interviewer.AI**	**Talently.ai**	**Jobylon**
Tech Industry Focus	100% tech-focused hiring	General hiring across industries	Specializes in tech hiring (engineers, AI trainers)	General hiring across industries	Tech hiring with AI interviews	General hiring across industries
AI Resume Screening & Feedback	AI-powered, tech-specific screening	General AI-driven assessments	No resume screening (focuses on expert vetting)	Maps resumes to job descriptions	No structured resume screening	No AI resume screening
Technical MCQ & Coding Tests	MCQs, text-based questions, auto-scored	Includes technical assessments but not industry-specific	No structured coding tests (focus on pre-vetted engineers)	Assesses technical and soft skills	No structured coding tests	No coding tests
AI-Powered Video Interviews	AI-simulated interviews for tech roles	Structured AI video interviews	No AI video interviews	AI-powered smart video interviews	AI conducts live, conversational interviews	No AI video interviews
Final Candidate Ranking (Tech-Specific Weightage)	Custom ranking (e.g., 60% tech, 10% resume, 30% interview)	No customizable ranking	No ranking system (focus on sourcing engineers)	Stack ranks candidates	No ranking system	No AI-driven ranking
AI-Powered Candidate Feedback	Yes, at all stages	No feedback system	No feedback system	No feedback system	No feedback system	No feedback system
High-Volume Tech Hiring	Automates screening & ranking for scale	Supports high-volume hiring but not tech-specific	No (focuses on hiring top-tier engineers only)	Automates pre-screening for large pools	No (focuses on live AI interviews)	Supports high-volume hiring, but not AI-driven
Freelance & Global Tech Hiring	Yes	No	Yes, hires and manages international engineers	No	No	Scalable for global hiring, but not AI-driven
Customization & Workflow Automation	Custom hiring stages for tech roles	Limited customization	No workflow automation (focus on pre-vetting)	Limited customization	No workflow automation	Strong customization & workflow automation

1.2 Problem Statement

The hiring process in the IT industry is plagued by inefficiencies:

• Time-Consuming Process: Companies manually review hundreds of resumes, leading to significant delays in identifying suitable candidates.

• Human Bias and Errors: The manual process introduces subjective biases, and candidates may misrepresent themselves through fake or misleading information on their resumes.

• Lack of Timely Feedback: Candidates often do not receive immediate feedback, leading to frustration and uncertainty in the recruitment pipeline.

• Resource Wastage: Companies waste valuable time and resources during interviews, which could be spent more productively.

This project aims to streamline the hiring process by automating key components of candidate evaluation through AI, offering both faster decisions and a better experience for candidates.

1.3 Scope

Our AI-driven interviewer streamlines recruitment by automating resume screening, test creation, and interview monitoring, ensuring a faster and fairer hiring process.

Designed for students, job seekers, and companies, it leverages AI for objective evaluation, real-time assessments, and customizable scoring.

With dedicated company and candidate portals, it simplifies job applications, talent acquisition, and decision-making, making hiring more efficient, transparent, and unbiased.

1.4 Modules

This section describes the key functional units or modules of the project, each designed to perform a specific task. The breakdown of the project into modules ensures that each part of the system is manageable and addresses specific objectives. Each module should be described in terms of its functionality and how it interacts with other modules in the system.

The modules should be logically organized to reflect the overall system workflow, ensuring that each module description is comprehensive but concise.

1.4.1 Module 1: AI Resume Screening System

Objective: Automate the evaluation of candidate resumes based on job requirements

Features:

1. **Resume Parsing Engine**
   - Convert diverse CV formats (PDF, DOCX, JPG) into standardized data structure
   - Extract key information (skills, experience, education, projects)
2. **Job-Resume Matching System**
   - Match extracted resume data against job description requirements
   - Calculate comprehensive match percentage and fit score
   - Identify key skills present and missing for the position
3. **Fraud Detection System**
   - Identify inconsistencies and potential misrepresentations
   - Flag suspicious patterns or unlikely qualifications
   - Validate claimed certifications and education when possible
4. **Resume Recommendation Engine**
   - Provide personalized improvement suggestions for candidates
   - Highlight critical missing skills for specific positions
   - Suggest format and content optimizations

1.4.2 Module 2: AI Assessment Generation System

Objective: Create tailored tests that accurately evaluate candidates' skills for specific positions

Features:

1. **Dynamic Question Generation**
   - AI-powered creation of job-specific questions
   - Support for multiple formats
2. **Test Configuration Interface**

- Customizable test parameters (duration, difficulty, question types)
- Template management for recurring positions
- Optional manual question addition and editing

3. **Progressive Assessment Engine**
   - Adaptive difficulty based on candidate performance
   - Skill-specific assessment sections
   - Time management and efficiency metrics
4. **Automated Scoring System**
   - Objective evaluation of technical responses
   - Natural language processing for text-based answers
   - Comparative scoring against benchmark responses

1.4.3 Module 3: AI Interview System

Objective: Conduct and analyze automated interviews to evaluate technical knowledge and soft skills

Features:

1. **Conversational Interview Interface**
   - Natural language dialogue management
   - Context-aware follow-up questions
   - Support for both text and voice-based interactions
2. **Technical Skill Evaluation**
   - Domain-specific knowledge assessment
   - Problem-solving capability measurement
   - Logical reasoning and analytical thinking evaluation
3. **Communication Analysis Engine**
   - Speech pattern and clarity assessment
   - Body language and eye contact analysis (video mode)
   - Confidence and presentation skills evaluation
4. **Real-time Monitoring Dashboard**

- Interview progress tracking for administrators
- Performance indicators during live sessions
- Technical issue detection and resolution

1.4.4 Module 4: Comprehensive Scoring and Analytics System

Objective: Provide data-driven evaluation and insights across the entire recruitment process

Features:

1. **Multi-dimensional Scoring Framework**
   - Weighted evaluation across all assessment stages
   - Customizable scoring priorities for different roles
   - Comparative ranking among candidate pool
2. **Performance Analytics Dashboard**
   - Visual representation of candidate strengths/weaknesses
   - Skill gap analysis with industry benchmarks
   - Historical performance tracking
3. **Hiring Process Optimization**
   - Recruitment funnel analytics
   - Predictive models for candidate success

1.4.5 Module 5: Integration and Platform Management

Objective: Ensure seamless operation and management of the entire system

Features:

1. **User Management System**
   - Role-based access control for different stakeholders
   - Customizable permissions and authentication
   - Profile management for candidates and company users
2. **Dual Portal Interface**

- Company portal for job posting and candidate management
- Candidate portal for applications and feedback

3. **Notification and Communication System**
   - Automated status updates and reminders
   - Feedback delivery to candidates
4. **API and Third-party Integration**
   - Integration with applicant tracking systems
   - Calendar synchronization for scheduling
   - Data export capabilities for external analysis

1.5 Work Division

Table 1.2: Work Divivsion

Name	Registration	Responsibility / Module / Feature
Hamza Saeed	21i-0671	Candidate Portal Frontend Resume Screening AI Scoring System Analytics Dashboard System Integration User Testing
Waleed Saeed	21i-1672	Data Preprocessing Resume Screening AI Test Creation AI Interview AI AI Integration Testing
Asad Ullah Shah	21i-0545	UI/UX Design Company Portal Frontend Test Creation AI Backend Development Cloud Deployment Testing

Chapter 2

Project Requirements

This chapter outlines the comprehensive requirements that guided the development of InterviewDen. All functional and non-functional requirements detailed in this section have been successfully implemented and validated through testing.

2.1 Use-case/Event Response Table/Storyboarding

1. **Upload and Optimize Resume**
   Allows candidates to upload their resume for AI analysis and improvement suggestions to increase match rates with relevant job positions.

2. **Search and Apply for Jobs**
   Enables candidates to browse available positions with filtering options and submit applications with personalized match scoring.

3. **Complete Skills Assessment**
   Facilitates customized technical and soft skills evaluation through various question formats tailored to specific job requirements.

4. **Participate in AI Interview**
   Provides automated preliminary interviews that evaluate technical knowledge, communication skills, and problem-solving abilities.

5. **Track Application Status**
   Offers candidates real-time visibility into their application progress across all stages with personalized feedback.

6. **Post Job Listing**
   Allows companies to create detailed job postings with specific skill requirements, assessment parameters, and application workflows.

7. **Configure AI Assessment**
   Enables employers to customize skill evaluation tests with specific difficulty levels, question types, and scoring criteria.

8. **Review Screened Candidates**
   Presents companies with AI-ranked applicants featuring match scores, skill alignment, and resume analysis findings.

9. **Analyze Test Results**
   Provides detailed analysis of candidate test performance with comparative metrics across key competencies.

10. **Review AI Interview Results**
    Delivers automated scoring of candidate interviews with analysis of technical accuracy, communication skills, and problem-solving.

11. **Generate Recruitment Analytics**
    Creates comprehensive metrics on hiring efficiency, time-to-hire, candidate quality, and process effectiveness.

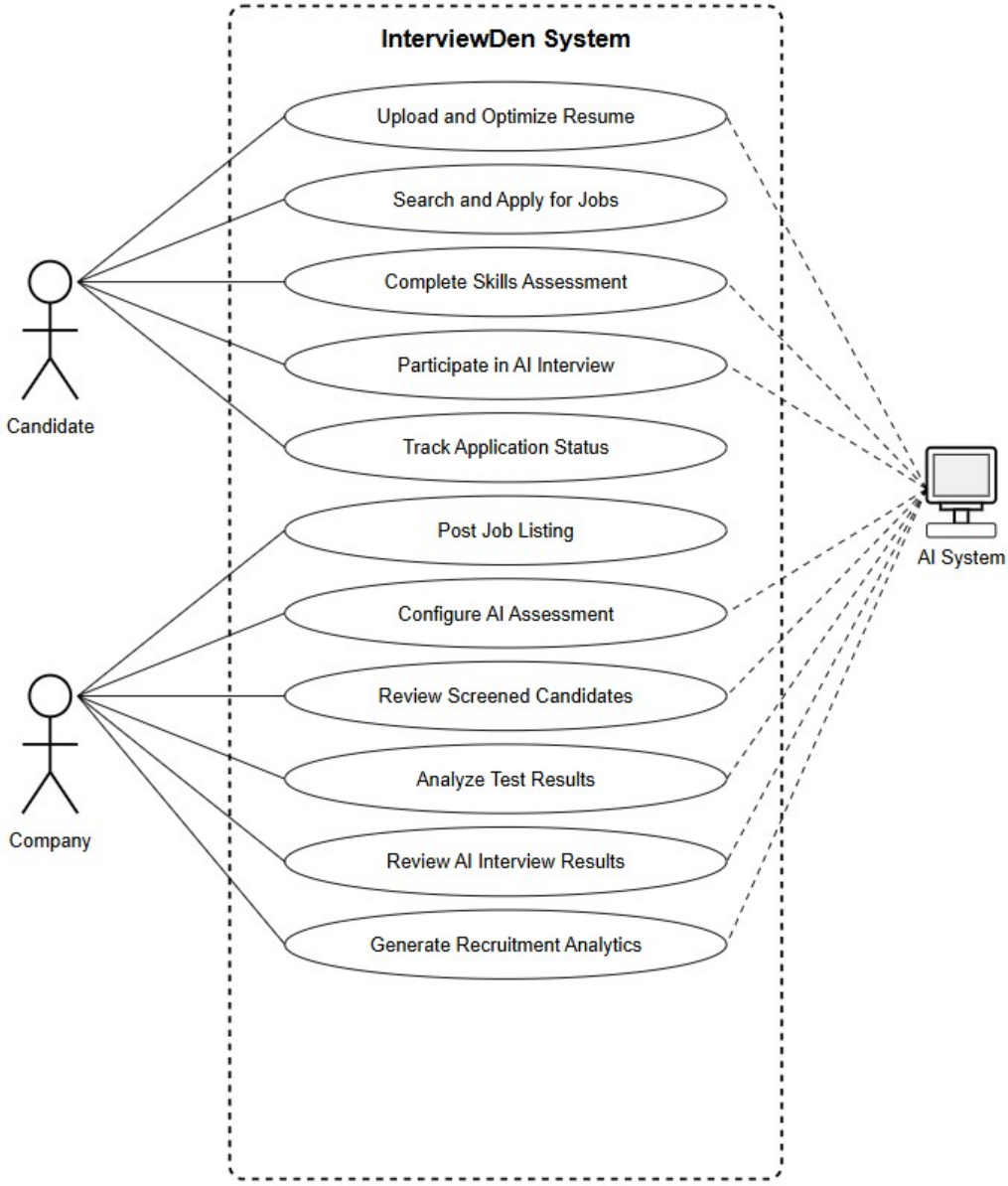


Figure 2.1: Use Case Diagram

2.2 Functional Requirements

2.2.1 Module 1

Following are the requirements for module 1:

1. FR-1.1: The system shall parse uploaded resumes with 95% accuracy for extracting key professional information.
2. FR-1.2: The system must support resume uploads in PDF and DOCX formats.
3. FR-1.3: The system shall generate optimization suggestions tailored to industry-specific standards.
4. FR-1.4: Resume processing and analysis must complete within 30 seconds of upload.
5. FR-1.5: The system shall perform multi-factor matching between resume skills and job requirements.
6. FR-1.6: The system must detect potential resume misrepresentations or inconsistencies.
7. FR-1.7: Matching algorithms shall consider at least 10 different professional criteria.

2.2.2 Module 2

Following are the requirements for module 2:

1. FR-2.1: The system shall generate customized assessments based on specific job requirements.
2. FR-2.2: Assessment generation must support multiple programming languages and question types.
3. FR-2.3: The system shall implement adaptive difficulty levels matching job seniority.
4. FR-2.4: Assessment question sets must be automatically updated quarterly with industry-relevant content.
5. FR-2.5: The system shall provide auto-save functionality during skills assessments.

6. FR-2.6: Assessment scoring must complete within 2 minutes of submission.
7. FR-2.7: The system must support multiple assessment formats including multiple-choice, coding, and text responses.
8. FR-2.8: The assessment system shall monitor and flag potential cheating behaviors.

2.2.3 Module 3

Following are the requirements for module 3:

1. FR-3.1: The system shall conduct automated technical and behavioral interviews.
2. FR-3.2: The interview system shall dynamically adapt questions based on previous candidate responses.
3. FR-3.3: Interview sessions must be recorded and stored securely for company review.
4. FR-3.4: Real-time analysis of candidate responses shall evaluate both technical accuracy and communication effectiveness.
5. FR-3.5: The system must maintain a natural conversation flow during automated interviews.
6. FR-3.6: Interview analysis must complete within 10 minutes of session completion.

2.2.4 Module 4

Following are the requirements for module 4:

1. FR-4.1: The system shall generate comprehensive recruitment funnel visualizations.
2. FR-4.2: The system shall analyze and score effectiveness of different recruitment screening stages.
3. FR-4.3: Recruitment analytics must refresh data at least hourly.
4. FR-4.4: The system shall generate process improvement recommendations based on historical data.
5. FR-4.5: The system must implement bias detection mechanisms in candidate evaluation.
6. FR-4.6: Performance benchmarking shall compare against industry standards.

2.2.5 Module 5

Following are the requirements for module 5:

1. FR-5.1: The system shall maintain user status updates within 1 hour of any change.
2. FR-5.2: Multiple notification channels must be supported, including email and in-app notifications.
3. FR-5.3: Critical system interactions must complete within specified performance thresholds:
   (a) Job search results: Load within 3 seconds
   (b) Candidate dashboard: Load within 2 seconds
   (c) Job listing publishing: Complete within 5 seconds
4. FR-5.4: All system modules must implement auto-save functionality.
5. FR-5.5: The platform shall ensure compatibility across multiple devices and browsers.
6. FR-5.6: User authentication and profile management must comply with data privacy regulations.
7. FR-5.7: The system shall implement secure mechanisms to prevent unauthorized access.

2.3 Non-Functional Requirements

2.3.1 Reliability

REL-1: The system shall maintain a Mean Time Between Failures (MTBF) of at least 720 hours (30 days) of continuous operation.

REL-2: In the event of a system failure, no more than 0.1% of ongoing candidate assessments or interviews shall be irretrievably lost.

REL-3: The system must implement comprehensive error logging and tracking mechanisms to capture all critical system failures.

REL-4: Automatic recovery procedures shall be initiated within 30 seconds of detecting a system anomaly.

REL-5: Critical data (user profiles, ongoing assessments, interview recordings) must be backed up in real-time with redundant storage.

REL-6: The system shall provide clear error messages that guide users or administrators on potential recovery actions.

2.3.2 Usability

USE-1: The user interface shall be navigable with no more than three clicks to access any primary system function.

USE-2: The system must support accessibility standards, including full keyboard navigation and screen reader compatibility.

USE-3: First-time users shall achieve basic task completion (resume upload, job application) with 90% success rate within 10 minutes of initial interaction.

USE-4: Context-sensitive help and tooltips must be available for all complex system interactions.

USE-5: The user interface shall maintain consistent design language across all modules, with intuitive iconography and layout.

USE-6: System shall support multiple languages, with at least 5 major international languages fully localized.

USE-7: Error messages must be written in clear, non-technical language that guides users toward resolution.

2.3.3 Performance

PER-1: 95% of all system interactions shall complete within 2 seconds under normal load conditions.

PER-2: The system must support concurrent operations for at least 10,000 active users without performance degradation.

PER-3: Resume parsing and optimization shall complete within 30 seconds for documents up to 10 pages long.

PER-4: Job search results must load within 3 seconds, displaying at least 50 relevant job listings.

PER-5: AI interview and assessment systems shall provide real-time response analysis with less than 500 milliseconds latency.

PER-6: Data visualization and analytics reports must generate within 5 seconds for datasets up to 10,000 records.

PER-7: System shall maintain consistent performance across desktop, tablet, and mobile platforms.

2.3.4 Security

SEC-1: The system must prevent unauthorized access with multi-factor authentication mechanisms.

SEC-2: All sensitive personal and professional data shall be encrypted at rest and in transit using industry-standard encryption protocols.

SEC-3: The system shall implement comprehensive audit logging for all critical user actions and administrative functions.

SEC-4: User authentication attempts shall be limited to 5 consecutive failed attempts before temporary account lockout.

SEC-5: Personal identification information must be protected with role-based access controls.

SEC-6: The system shall automatically detect and flag suspicious user behavior patterns.

SEC-7: Compliance with GDPR, CCPA, and other relevant data protection regulations must be maintained.

SEC-8: All system integrations and third-party data exchanges must use secure, authenticated API connections.

2.3.5 Scalability

SCA-1: The system architecture must support horizontal scaling to accommodate 50% user base growth without performance degradation.

SCA-2: Database operations shall maintain consistent performance with data volumes increasing by up to 100% annually.

SCA-3: Microservices architecture must allow independent scaling of individual system components.

2.3.6 Compatibility

COM-1: The system shall be fully functional on the following platforms:

- Web browsers: Chrome, Firefox, Safari, Edge (latest two versions)
- Desktop operating systems: Windows, macOS, Linux

2.3.7 Maintainability

MAI-1: System code shall maintain a documentation coverage of 90% for all critical modules.

MAI-2: Automated testing must provide code coverage of at least 85% across all system components.

MAI-3: System updates and patches must be deployable with zero downtime.

Chapter 3

System Overview

InterviewDen successfully addresses critical recruitment challenges through an implemented AI-powered platform. This chapter presents the system architecture, data design, and design models that guided the development and now form the foundation of the production system. The platform transforms talent acquisition by providing advanced capabilities for both organizations and job seekers. For candidates, the system offers sophisticated profile optimization and precise job matching beyond traditional screening methods. Organizations benefit from a streamlined workflow that delivers comprehensive insights into candidate potential, reducing hiring cycle times and improving selection quality. By implementing advanced analytical methodologies, InterviewDen mitigates traditional hiring biases and provides a holistic assessment of professional capabilities. The platform represents a strategic approach to connecting organizational needs with professional talent through intelligent, data-driven solutions.

3.1 Architectural Design

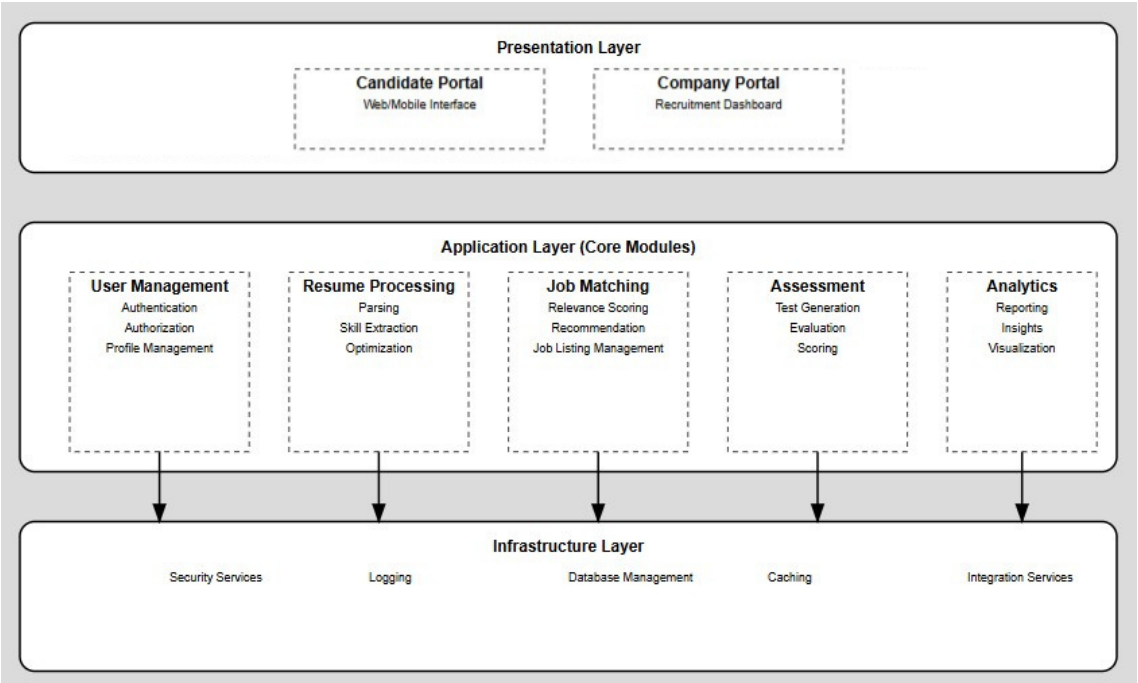


Figure 3.1: System Architecture

The system architecture diagram illustrates InterviewDen's comprehensive technology platform, organized into three distinct layers. At the top, the Presentation Layer provides user interfaces for both candidates and companies, serving as the primary access points to the system. The middle Application Layer constitutes the platform's core functionality, comprising five integrated modules: User Management, Resume Processing, Job Matching, Assessment, and Analytics. These modules feature specialized capabilities that collaborate seamlessly to optimize the recruitment workflow. The foundational Infrastructure Layer provides critical support services including security mechanisms, logging systems, database management, caching solutions, and integration capabilities. This layered architecture enables a flexible, scalable, and efficient recruitment technology solution capable of adapting to varied hiring requirements while ensuring consistent performance and seamless component interaction.

3.2 Design Models

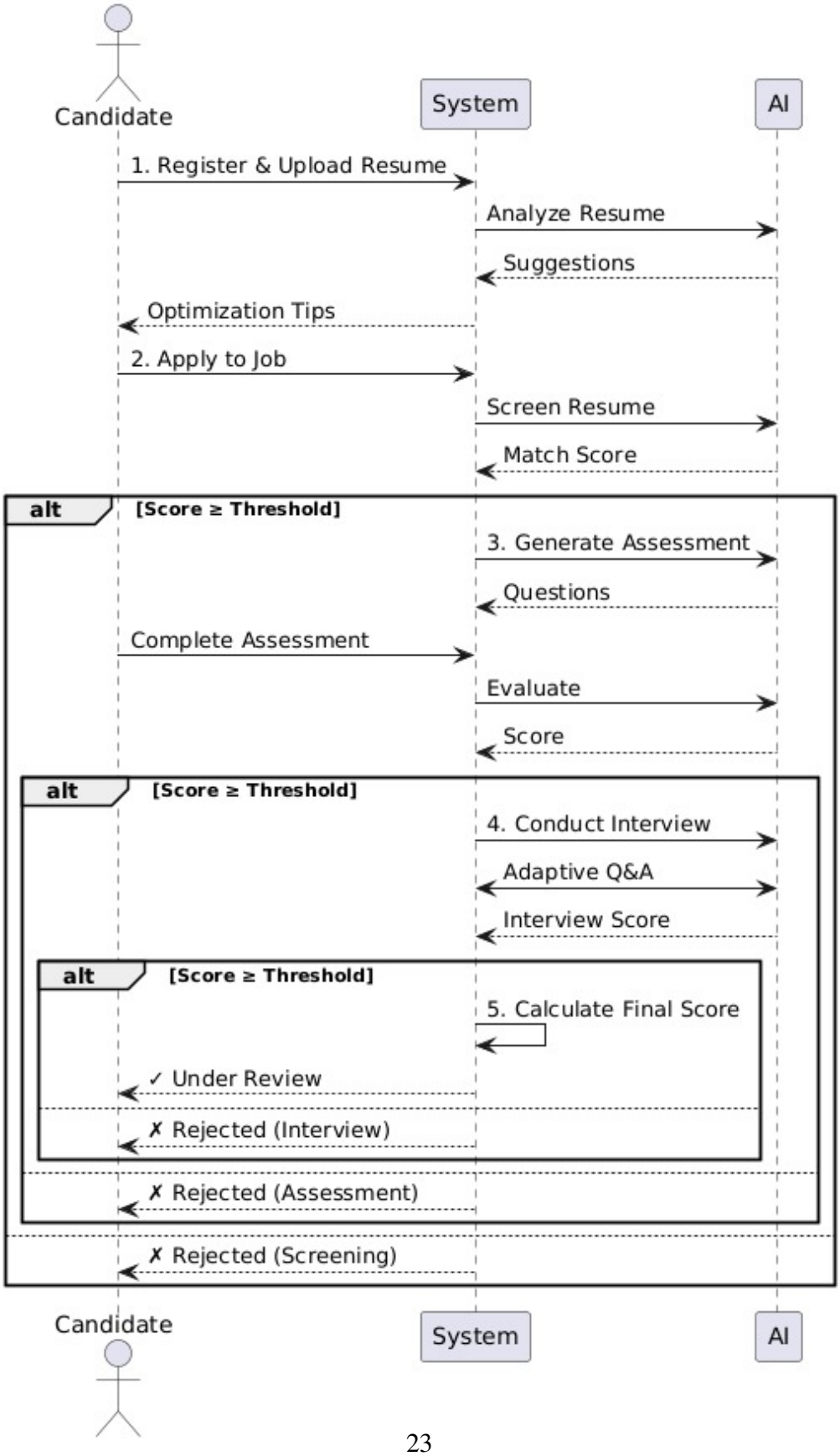


Figure 3.2: State Diagram - Application Status Transitions

The application progresses through the following states:

- **PENDING**: Initial state when candidate submits application
- **SCREENING**: AI-powered resume screening using fine-tuned Llama 8 model
- **ASSESSMENT**: Candidate completes skills assessment with LLM-generated questions
- **INTERVIEW**: AI-conducted adaptive interview using Groq API
- **UNDER_REVIEW**: Final evaluation by company after passing all stages
- **ACCEPTED/REJECTED**: Terminal states based on evaluation results

3.3 Data Design

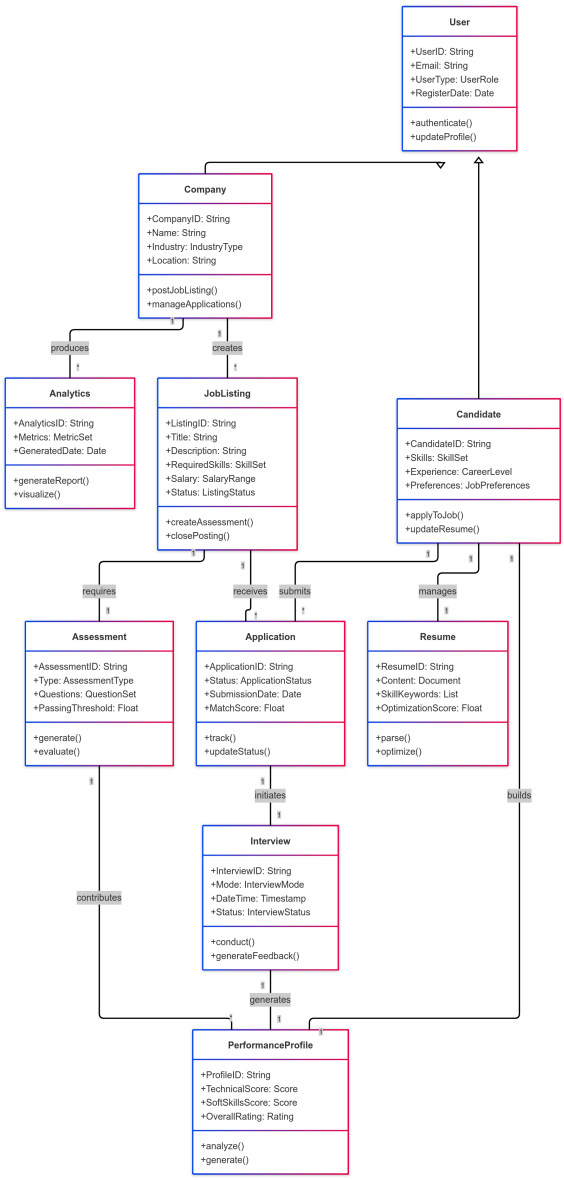


Figure 3.3: Data Design Diagram

3.4 Domain Model

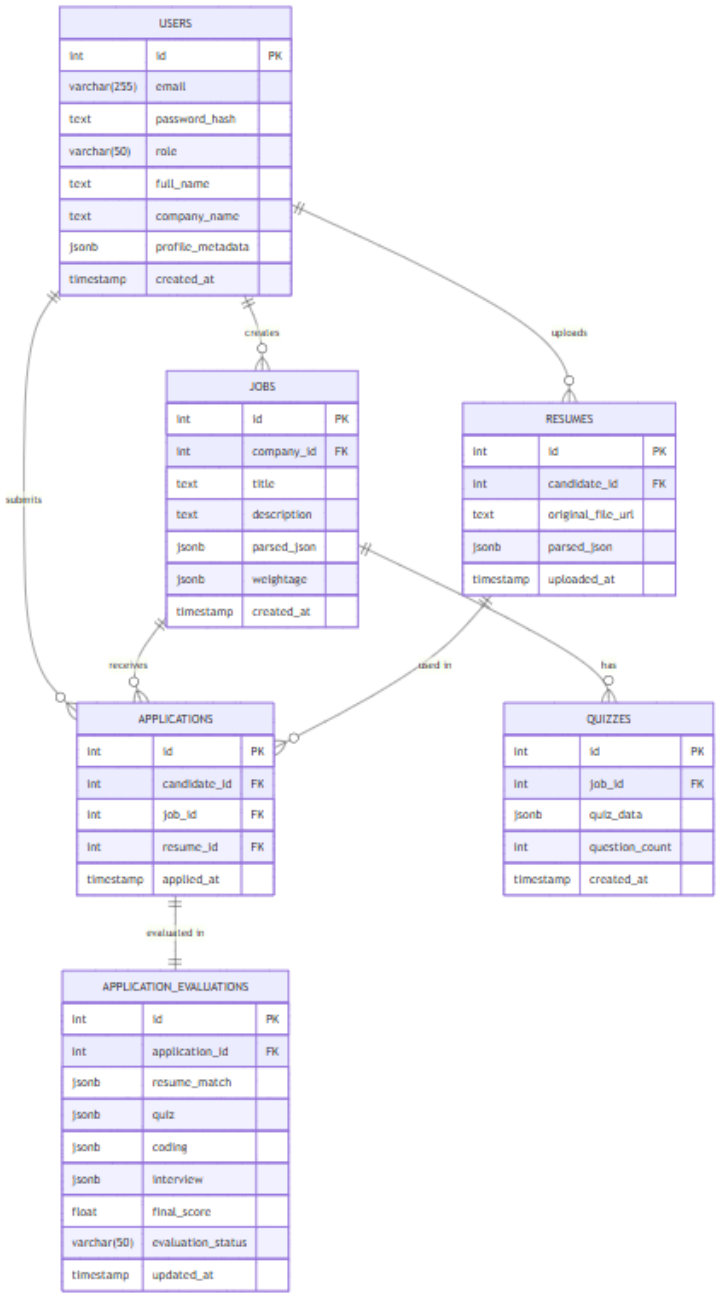


Figure 3.4: Domain Model Diagram

Chapter 4

Implementation and Testing

4.0.1 System Context and Functionality

The AI-Driven Job Interview Analyzer addresses critical inefficiencies in modern recruitment by automating three core hiring processes: resume screening, skill assessment through MCQ tests, and AI-conducted interviews. The system operates through a dual-portal architecture where companies create job-specific evaluation criteria and candidates undergo comprehensive automated assessment.

The platform leverages artificial intelligence at multiple levels—from fine-tuned machine learning models for resume analysis to large language model integration for contextual evaluation and conversational interviews. By combining these technologies with interactive avatar-based interviewing through Akool.ai API integration, the system provides an end-to-end automated hiring solution that maintains human-like interaction quality while eliminating manual bottlenecks.

4.0.2 Technical Architecture

The system comprises three primary modules operating in sequential workflow:

1. **Resume Screening Module**: Employs a fine-tuned machine learning model to automatically filter and rank candidates based on alignment between their qualifications and job requirements.

2. **MCQ Test Generation and Evaluation Module**: Utilizes LLM-based contextual analysis to generate job-specific questions and evaluate candidate responses with semantic understanding.

3. **AI Interview Module**: Conducts interactive interviews using LLM for conversational intelligence and Akool.ai API for realistic avatar presentation, providing

comprehensive behavioral and technical assessment.

Each module produces structured output that feeds into subsequent stages, creating a comprehensive candidate evaluation pipeline that maintains consistency while reducing human intervention requirements.

4.1 Algorithm Design

4.1.1 Module 1: Resume Screening Algorithm

4.1.1.1 Overview and Approach

The resume screening module employs a fine-tuned machine learning model specifically trained to match candidate qualifications with job requirements.

4.1.1.2 Model Architecture

We fine-tuned a Llama8 model on a curated dataset of resume-job description pairs with labeled match scores. The model learns to:

- Extract semantic meaning from unstructured resume text
- Identify relevant skills and experience indicators
- Recognize skill equivalencies and related competencies
- Weight different qualification aspects based on job requirements

4.1.1.3 Fine-Tuning Process

The model fine-tuning involved:

1. **Dataset Preparation**: Collected resume-job pairs with expert-labeled match scores
2. **Data Augmentation**: Applied synonym replacement and paraphrasing to increase robustness

4.1.1.4 Advantages of Fine-Tuned Approach

- **Domain Adaptation**: Model learns industry-specific terminology and qualification patterns
- **Implicit Skill Relations**: Captures relationships between related skills without explicit rules
- **Consistent Evaluation**: Provides reproducible scores based on learned patterns from successful hires
- **Scalability**: Processes hundreds of resumes in minutes without degradation in accuracy

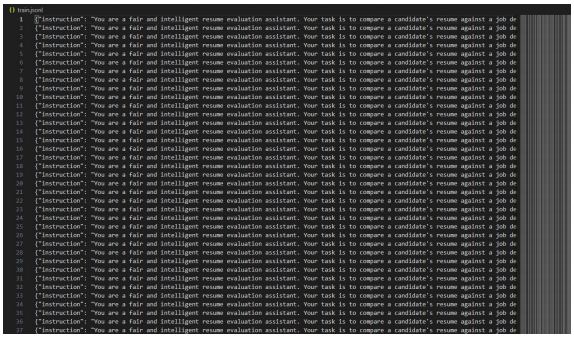


Figure 4.1: Data For Finetuning

```python
from trl import SFTTrainer
from transformers import TrainingArguments
from unsloth import is_bfloat16_supported

trainer = SFTTrainer(
    model = model,
    tokenizer = tokenizer,
    train_dataset = dataset,
    dataset_text_field = "text",
    max_seq_length = max_seq_length,
    dataset_num_proc = 2,
    args = TrainingArguments(
        per_device_train_batch_size = 2,
        gradient_accumulation_steps = 4,
        warmup_steps = 5,
        max_steps = 60,
        learning_rate = 2e-4,
        fp16 = not is_bfloat16_supported(),
        bf16 = is_bfloat16_supported(),
        logging_steps = 1,
        optim = "adamw_8bit",
        weight_decay = 0.01,
        lr_scheduler_type = "linear",
        seed = 3407,
        output_dir = "outputs",

        # --- ADD THIS LINE TO DISABLE WANDB ---
        report_to = "none",
    ),
)

trainer.train()
```

Figure 4.2: Finetuning For Resume Screening

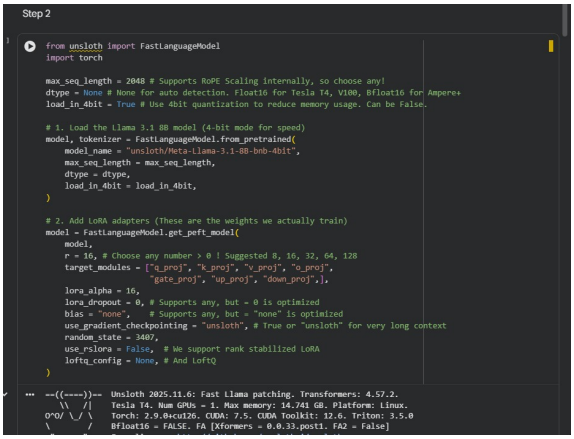


Figure 4.3: Finetuning For Resume Screening

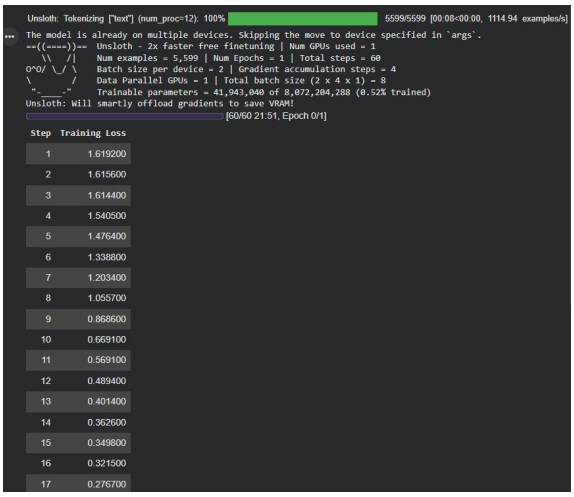


Figure 4.4: Training

4.1.2 Module 2: MCQ Test Generation and Evaluation

4.1.2.1 Overview and Approach

The MCQ test module leverages large language models for contextual generation of job-specific questions and intelligent evaluation of candidate responses. Unlike static question banks, this approach generates tailored questions that assess actual understanding rather than memorization, adapting difficulty based on job requirements.

4.1.2.2 LLM-Based Contextual Analysis

The system uses prompt engineering to guide the LLM in:

- Generating questions aligned with specific job skills and difficulty levels
- Creating plausible distractors that test understanding depth

4.1.2.3 Prompt Engineering Strategy

To ensure consistent and high-quality question generation, we employ:

1. **Structured Output Format**: Prompts specify exact JSON schema for LLM responses
2. **Constraint Specification**: Explicitly state requirements (e.g., "avoid ambiguous wording")
3. **Validation Instructions**: Request self-verification within the prompt

```python
agent_generator.py
Part2 > agent_generator.py > ...
class QuestionGeneratorAgent:
    """Generates multiple-choice technical questions."""

    def __init__(self):
        api_key = config.get_current_api_key()
        self.llm = ChatGroq(api_key=api_key, model_name=config.MODEL_NAME)
        print("✅ QuestionGeneratorAgent initialized.")

    def generate_questions(self, job_data: Dict[str, Any], num_questions: int) -> List[Dict[str, Any]]:
        job_title = job_data.get("job_title", "Unknown Title")
        skills = ", ".join(job_data.get("skills_required", {}).get("technical_skills", []))
        responsibilities = ", ".join(job_data.get("job_responsibilities", []))
        experience = job_data.get("experience_required", {}).get("years_of_experience", "Not specified")
        level = job_data.get("experience_required", {}).get("level", "Not specified")

        # Define difficulty descriptions
        level_descriptions = {
            "Internship": "Focus on fundamental concepts and basic skills appropriate for someone new to the field",
            "Associate": "Cover core skills with some practical application for early career professionals",
            "Junior": "Include practical scenarios and common problem-solving for developers with some experience",
            "Senior": "Emphasize complex scenarios, architecture decisions, and advanced concepts for experienced professionals",
            "Expert": "Focus on cutting-edge technologies, system design, and expert-level problem-solving for industry leaders"
        }

        level_description = level_descriptions.get(level, "Appropriate for the role's requirements")

        prompt_text = (
            f"You are an expert technical interviewer.\n\n"
            f"Job Title: {job_title}\n"
            f"Skills: {skills}\n"
            f"Responsibilities: {responsibilities}\n"
            f"Experience Level: {level} - {level_description}\n"
            f"Years of Experience: {experience}\n\n"
            f"Generate {num_questions} concise, technically accurate multiple-choice questions (MCQs) "
            f"appropriate for a {level} level candidate.\n"
            f"The questions should match the difficulty level:\n"
```

Figure 4.5: Test Generator

4.1.2.4 Advantages of LLM-Based Approach

- **Dynamic Question Generation**: Creates unique questions for each test, preventing memorization
- **Context-Aware Assessment**: Evaluates understanding within the specific job context
- **Intelligent Feedback**: Provides explanatory feedback that aids candidate learning
- **Adaptive Difficulty**: Adjusts question complexity based on role seniority

```
test_generator.py
Part2 > test_generator.py > ...
31  class JobTestGenerator:
38      def generate_batch_questions(self, job_title, skills, responsibilities, experience, level, num_questions):
51          level_description = level_descriptions.get(level, "Appropriate for the role's requirements")
52
53          prompt = f"""
54  You are a technical interviewer.
55  Generate {num_questions} multiple-choice questions (MCQs) for the following role.
56
57  Job Title: {job_title}
58  Skills: {skills_str}
59  Responsibilities: {resp_str}
60  Experience: {experience}
61  Level: {level} - {level_description}
62
63  The questions should match the difficulty level:
64  - Internship: Basic concepts, fundamental knowledge
65  - Associate: Core skills, simple applications
66  - Junior: Practical scenarios, common problem-solving
67  - Senior: Complex scenarios, architecture decisions
68  - Expert: Advanced concepts, system design, cutting-edge technologies
69
70  Each question must have:
71  - Four options labeled A, B, C, D
72  - One correct answer clearly indicated as "Correct answer: X"
73  - Appropriate difficulty for a {level} level candidate
74
75  Use this exact format:
76
77  1. Question text
78  A. Option A
79  B. Option B
80  C. Option C
81  D. Option D
82  Correct answer: A
83  """
```

Figure 4.6: Test Instructions

4.1.3 Module 3: AI Interview Conductor

4.1.3.1 Overview and Approach

The AI interview module combines LLM-based conversational intelligence with Akool.ai avatar API to conduct realistic, interactive interviews. The system maintains natural conversation flow while systematically assessing technical knowledge, problem-solving ability, and communication skills through multi-turn dialogue.

4.1.3.2 Integration Architecture

The module integrates two primary technologies:

- **LLM for Conversation Management**: Generates contextually appropriate questions, interprets candidate responses, and adapts interview direction based on answers
- **Akool.ai Avatar API**: Provides realistic visual representation of the interviewer with synchronized speech and expressions, enhancing candidate engagement

4.1.3.3 Akool.ai Integration Details

The avatar integration enhances interview realism through:

1. **Text-to-Speech Synthesis**: Converts generated questions to natural-sounding speech
2. **Facial Animation**: Synchronizes avatar expressions with speech for natural presentation
3. **Real-time Rendering**: Generates and displays avatar video with minimal latency

4.1.3.4 Conversational Intelligence

The LLM manages interview flow through:

- **Dynamic Question Selection**: Adapts questions based on previous responses
- **Follow-up Generation**: Asks clarifying questions when responses are incomplete
- **Topic Coverage**: Ensures all required skills are assessed during the interview
- **Natural Transitions**: Maintains conversational flow between different topics

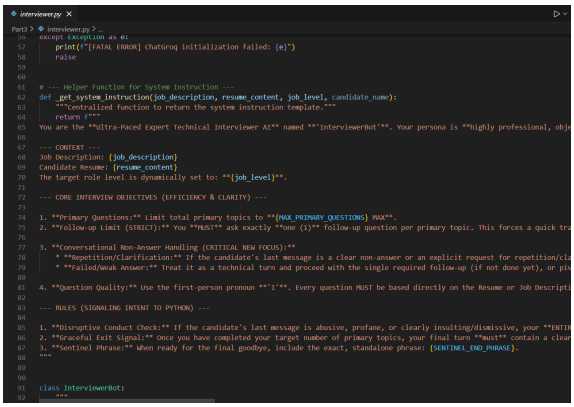
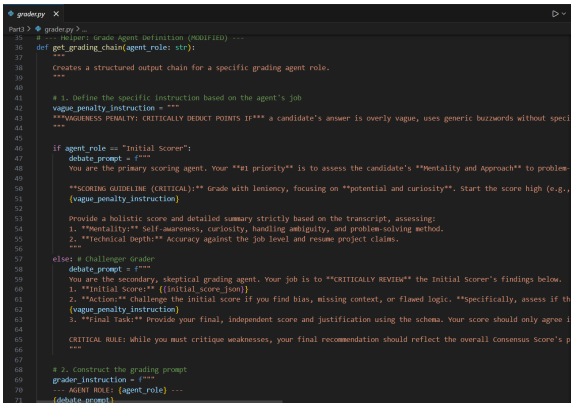


Figure 4.7: Interview Instructions

4.1.3.5 Advantages of AI Interview Approach

- **Consistent Evaluation**: All candidates face standardized but adaptive assessment
- **Bias Reduction**: Eliminates unconscious biases related to appearance or background
- **Scalability**: Conducts multiple interviews simultaneously without quality degradation
- **Comprehensive Recording**: Captures complete transcript for detailed analysis
- **Candidate Comfort**: Some candidates feel more comfortable with AI than human interviewers



```
grader.py
Part3 > grader.py > ...
# --- Ma'ayan: Grade Agent Definition (MODIFIED) ---
def get_grading_chain(agent_role: str):
    """
    Creates a structured output chain for a specific grading agent role.
    """

    # 1. Define the specific instruction based on the agent's job
    vague_penalty_instruction = """
    ***VAGUENESS PENALTY: CRITICALLY DEDUCT POINTS IF*** a candidate's answer is overly vague, uses generic buzzwords without speci
    """

    if agent_role == "Initial Scorer":
        debate_prompt = f"""
        You are the primary scoring agent. Your **#1 priority** is to assess the candidate's **Mentality and Approach** to problem-

        **SCORING GUIDELINE (CRITICAL):** Grade with leniency, focusing on **potential and curiosity**. Start the score high (e.g.,
        {vague_penalty_instruction}

        Provide a holistic score and detailed summary strictly based on the transcript, assessing:
        1. **Mentality:** Self-awareness, curiosity, handling ambiguity, and problem-solving method.
        2. **Technical Depth:** Accuracy against the job level and resume project claims.
        """
    else: # Challenger Grader
        debate_prompt = f"""
        You are the secondary, skeptical grading agent. Your job is to **CRITICALLY REVIEW** the Initial Scorer's findings below.
        1. **Initial Score:** {{initial_score_json}}
        2. **Action:** Challenge the initial score if you find bias, missing context, or flawed logic. **Specifically, assess if th
        {vague_penalty_instruction}
        3. **Final Task:** Provide your final, independent score and justification using the schema. Your score should only agree i

        CRITICAL RULE: While you must critique weaknesses, your final recommendation should reflect the overall Consensus Score's p
        """

    # 2. Construct the grading prompt
    grader_instruction = f"""
    --- AGENT ROLE: {agent_role} ---
    {debate_prompt}
```

Figure 4.8: Interview Grading

4.2 External APIs/SDKs

Table 4.1: External APIs and SDKs

API and Version	Description	Purpose of Usage	API Endpoint/ Function Used
Groq API	High-performance LLM inference platform	Assessment question generation, interview analysis	`api.groq.com/openai/v1/chat/completions`
Hugging Face (v4.35.0)	Open-source library for NLP models	Loading fine-tuned Llama 8 model, tokenization	`AutoModelForCausalLM`, `AutoTokenizer`
Akool.ai API	AI-powered avatar and video generation	Creating AI interviewer avatars, video feedback	`api.akool.com/v1/avatar/create`
Web Speech API	Browser-based speech recognition	Voice interview responses, text-to-speech	`SpeechRecognition()`, `speak()`
Supabase JS Client (v2.38.4)	Backend-as-a-Service client library	Database ops, auth, storage, realtime	`createClient()`, `from().select()`
Next.js (v14.0)	React framework with SSR	Full-stack framework, API routes	`app/`, `api/`, `middleware`
FastAPI (v0.104.1)	Modern Python web framework	AI microservice API server	`@app.post()`, `@app.get()`
PyTorch (v2.1.0)	Deep learning framework	Model inference for Llama 8	`torch.nn`, `model.generate()`
QLORA (v0.6.0)	Parameter-efficient fine-tuning	Fine-tuning Llama 8 model	`LoraConfig()`, `get_peft_model()`
Docker	Containerization platform	Application containerization	`Dockerfile`, `docker-compose`
Railway CLI	Cloud deployment platform	Deployment, monitoring	`railway up`, `railway logs`

4.3 Testing Details

4.3.1 Unit Testing

Individual components tested in isolation:

- Resume parser accuracy on diverse document formats
- Question generation quality and appropriateness
- Response analysis correctness and consistency

- Avatar API integration reliability

4.3.2 Integration Testing

Testing module interactions:

- Data flow between resume screening and test generation
- Candidate profile transfer from testing to interview modules
- Complete candidate journey from application to final assessment

4.3.3 System Testing

End-to-end testing of complete workflows:

- Company creates job posting and test requirements
- Candidate applies, takes test, and completes interview
- Reports generated and accessible to both parties
- System behavior under load with multiple concurrent users

Chapter 5

Conclusions and Future Work

The AI-Driven Job Interview Analyzer (Interview Den) provides an innovative solution to the inefficiencies in traditional hiring practices. By automating the resume screening, test generation, and interview processes using AI, the platform ensures faster, more accurate, and data-driven hiring decisions. The system reduces human biases, streamlines the recruitment process, and provides a better overall experience for both companies and candidates. This AI-powered platform has the potential to significantly enhance recruitment efficiency and candidate satisfaction in the IT industry and beyond.

Bibliography

Appendix A

Appendices

A.1 Appendix A

A.1.1 Detailed Use Case Diagrams

Building upon the use cases listed in the previous figure, these are the detailed descriptions for each use case.

Use Case 1

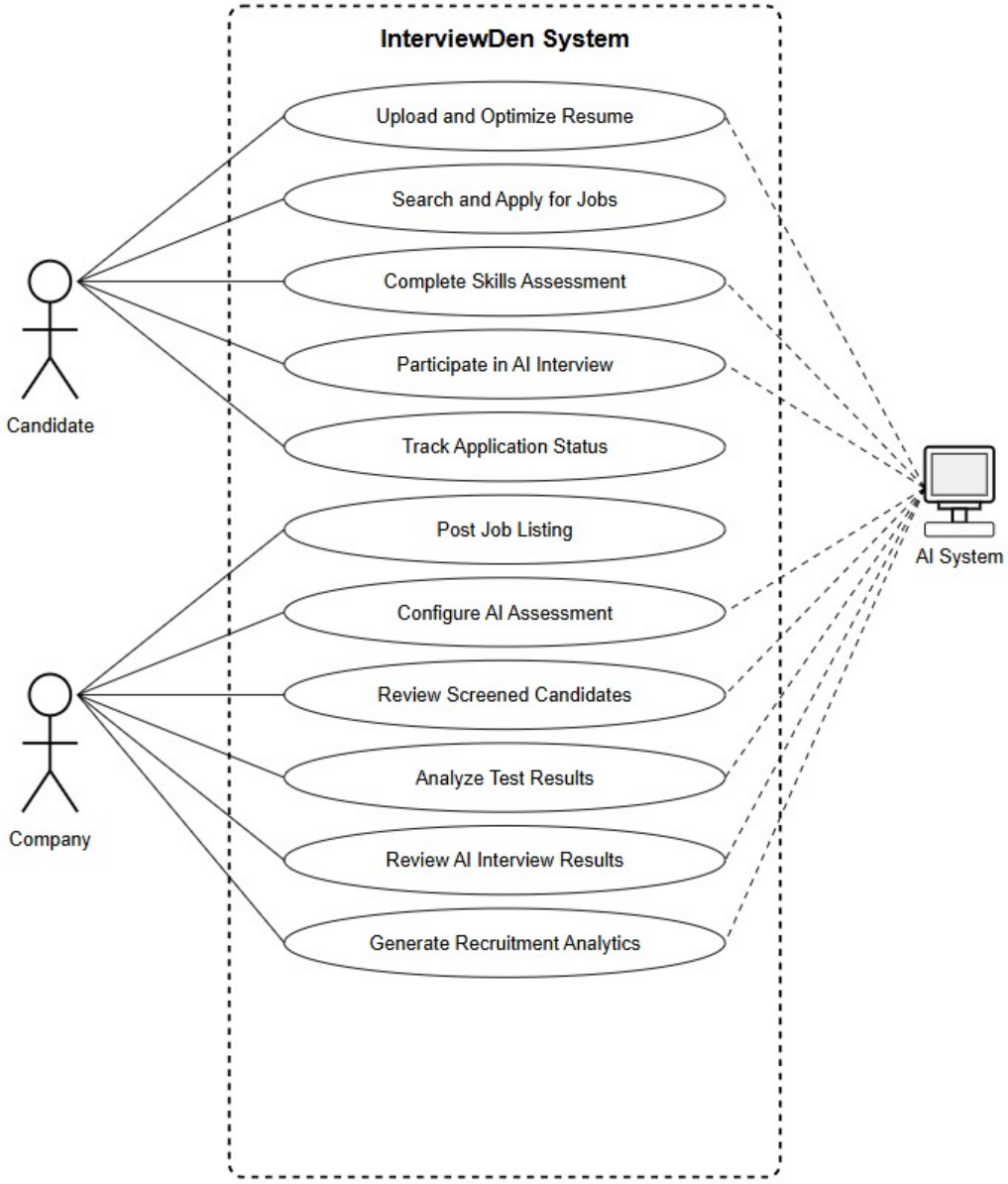


Figure A.1: Use Case Diagram

Use Case	**Upload and Optimize Resume**
Primary Actor	Candidate
Stakeholders	Candidate, AI System
Preconditions	Candidate registered and logged in
Main Success Scenario	1. Upload resume 2. Parse resume information 3. Analyze resume content 4. Present parsed information 5. Confirm or edit information 6. Provide optimization suggestions 7. Apply suggested changes 8. Save optimized resume
Extensions	• Unsupported file format • Parsing fails • Resume well-optimized
Special Requirements	• Resume parsing must maintain 95% accuracy for key fields • Optimization suggestions must be tailored to industry standards • Process must complete within 30 seconds

Table A.1: Use Case 1: Upload and Optimize Resume

Use Case 2

Use Case	**Search and Apply for Jobs**
Primary Actor	Candidate
Stakeholders	Candidate, Company
Preconditions	Profile created and resume uploaded
Main Success Scenario	1. Access job search 2. Enter search criteria 3. Display matching jobs 4. Filter results 5. Select job listing 6. View job details 7. Apply to job 8. Pre-fill application 9. Confirm application 10. Submit application
Extensions	• No matching jobs • Low match percentage • Missing profile information
Special Requirements	• Search results must load within 3 seconds • Match percentage calculation must consider at least 10 different factors

Table A.2: Use Case 2: Search and Apply for Jobs

Use Case 3

Use Case	**Complete Skills Assessment**
Primary Actor	Candidate
Stakeholders	Candidate, Company, AI System
Preconditions	Invited to assessment
Main Success Scenario	1. Receive assessment invitation 2. Schedule assessment 3. Start assessment 4. Complete various question types 5. System monitors for cheating 6. Evaluate code execution 7. Analyze response quality 8. Generate skills profile 9. Receive performance feedback 10. Company notified
Extensions	• Rescheduling request • Cheating detected • Code execution timeout
Special Requirements	• Assessment environment must support multiple programming languages • System must handle network interruptions with auto-save functionality • Assessment scoring must complete within 2 minutes of submission

Table A.3: Use Case 3: Complete Skills Assessment

Use Case 4

Use Case	**Participate in AI Interview**
Primary Actor	Candidate
Stakeholders	Candidate, Company, AI System
Preconditions	Passed initial screening
Main Success Scenario	1. Receive AI interview invitation 2. Schedule interview 3. Join interview environment 4. Receive interview process overview 5. Respond to technical and behavioral questions 6. AI analyzes responses 7. AI adapts questioning 8. Maintain conversation flow 9. Complete interview 10. Receive performance feedback
Extensions	• Technical communication issues • Unclear responses • Need for question clarification
Special Requirements	• Speech recognition accuracy must exceed 95% • System must handle accents and non-native speakers • Interview session must record and store for company review

Table A.4: Use Case 4: Participate in AI Interview

Use Case 5

Use Case	**Track Application Status**
Primary Actor	Candidate
Stakeholders	Candidate, Company
Preconditions	Submitted at least one application
Main Success Scenario	1. Access application dashboard 2. View application statuses 3. Select specific application 4. View application progress 5. Review stage feedback 6. Check upcoming steps 7. Receive improvement recommendations 8. Get status change notifications
Extensions	• Application rejected • Low assessment scores • No recent updates
Special Requirements	• Status updates must be reflected within 1 hour of company changes • Notification system must support email, SMS, and in-app alerts

Table A.5: Use Case 5: Track Application Status

Use Case 6

Use Case	**Post Job Listing**
Primary Actor	Company
Stakeholders	Company, Candidates
Preconditions	Company profile completed
Main Success Scenario	1. Access job creation interface 2. Enter job details 3. Define required skills 4. Set compensation information 5. Receive job description suggestions 6. Configure screening criteria 7. Set up assessment parameters 8. Define application workflow 9. Preview job listing 10. Publish listing 11. System indexes job
Extensions	• Potential bias detection • Incomplete assessment configuration • Save as draft
Special Requirements	• Job listing creation form must auto-save every 60 seconds • Publishing process must complete within 5 seconds

Table A.6: Use Case 6: Post Job Listing

Use Case 7

Use Case	**Configure AI Assessment**
Primary Actor	Company
Stakeholders	Company, Candidates, AI System
Preconditions	Job listing created
Main Success Scenario	1. Access assessment configuration 2. Select assessment types 3. Define technical skills to test 4. Generate suggested questions 5. Review and customize questions 6. Set difficulty levels 7. Configure time limits and scoring 8. Add custom questions 9. Validate assessment 10. Activate assessment 11. Link to job workflow
Extensions	• Insufficient job details • Assessment imbalance detected • Save as template
Special Requirements	• Question bank must be regularly updated with industry-relevant content • Assessment configuration must support multiple languages • System must ensure assessment difficulty matches job seniority level

Table A.7: Use Case 7: Configure AI Assessment

Use Case 8

Use Case	**Review Screened Candidates**
Primary Actor	Company
Stakeholders	Company, Candidates
Preconditions	Candidates applied for position
Main Success Scenario	1. Access applicant dashboard 2. View candidates by match percentage 3. Filter candidates 4. Select candidate for detailed review 5. Display comprehensive profile 6. Highlight matching skills 7. Identify potential inconsistencies 8. Review AI candidate evaluation 9. Make hiring decision 10. Update candidate status
Extensions	• No qualified candidates • Resume verification needed • Bulk candidate actions
Special Requirements	• Dashboard must load initial candidate list within 2 seconds • AI screening must achieve 90% agreement with human recruiters • System must comply with anti-discrimination regulations

Table A.8: Use Case 8: Review Screened Candidates

Use Case 9

Use Case	**Analyze Test Results**
Primary Actor	Company
Stakeholders	Company, Candidates, AI System
Preconditions	Candidates completed assessments
Main Success Scenario	1. Access test results dashboard 2. View overall performance metrics 3. Filter candidates 4. View comparative analysis 5. Select candidate for detailed review 6. Display test performance breakdown 7. Highlight strengths and weaknesses 8. Review code quality metrics 9. Analyze specific answers 10. Make hiring decision 11. Update candidate status
Extensions	• Unusual test patterns • Code plagiarism detected • Disagree with AI scoring
Special Requirements	• Performance analysis must include benchmarking against industry averages • System must analyze response patterns for insight into problem-solving approaches • Results must be visualized with clear graphics for quick understanding

Table A.9: Use Case 9: Analyze Test Results

Use Case 10

Use Case	**Review AI Interview Results**
Primary Actor	Company
Stakeholders	Company, Candidates, AI System
Preconditions	Candidates completed AI interviews
Main Success Scenario	1. Access interview results dashboard 2. View candidates by interview performance 3. Select specific candidate 4. Review interview transcript 5. Analyze technical accuracy 6. Assess communication effectiveness 7. Identify behavioral indicators 8. Replay interview segments 9. Review AI interview scoring 10. Make hiring decision 11. Update candidate status
Extensions	• Technical interview issues • Conflicting behavioral indicators • Disagree with AI assessment
Special Requirements	• Video playback must support variable speed and closed captioning • Interview analysis must be available within 10 minutes of completion • System must detect and flag potential biases in AI evaluation

Table A.10: Use Case 10: Review AI Interview Results

Use Case 11

Use Case	**Generate Recruitment Analytics**
Primary Actor	Company
Stakeholders	Company, AI System
Preconditions	Company has active or completed recruitment processes
Main Success Scenario	1. Company administrator accesses analytics dashboard 2. System displays high-level recruitment metrics 3. Administrator selects specific time period for analysis 4. System generates comprehensive recruitment funnel visualization 5. System provides metrics on time-to-hire and cost-per-hire 6. System analyzes effectiveness of different screening stages 7. System identifies bottlenecks in recruitment process 8. System compares current metrics against historical benchmarks 9. Administrator can drill down into specific positions or departments 10. System generates recommendations for process improvement 11. Administrator can export reports in various formats
Extensions	• Insufficient data: System notes limitations of analysis • Critical bottleneck identified: System highlights urgent attention needed • Administrator requests custom analysis: System provides configuration options
Special Requirements	• Dashboard must refresh data at least hourly • Analytics must be available for individual positions and aggregate across company • System must maintain data privacy in accordance with regulations

Table A.11: Use Case 11: Generate Recruitment Analytics

A.2 Appendix B

A.2.1 Domain Model Example For Online Shopping

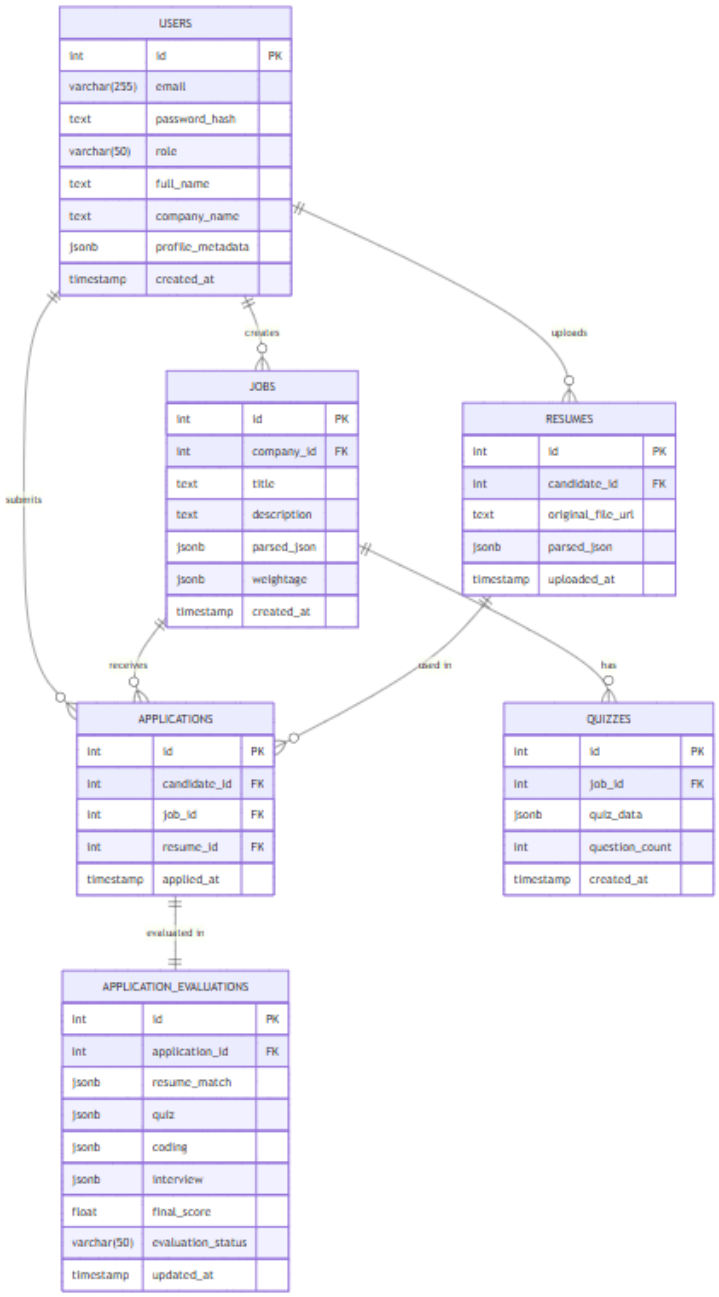


Figure A.2: Domain Model Diagram

A.3 Appendix C

A.3.1 System Sequence Diagram

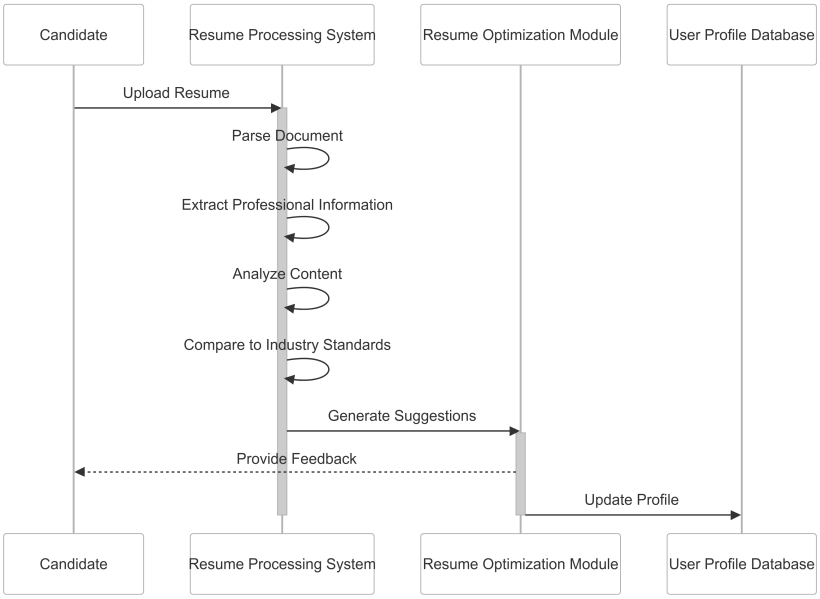


Figure A.3: SSD 1

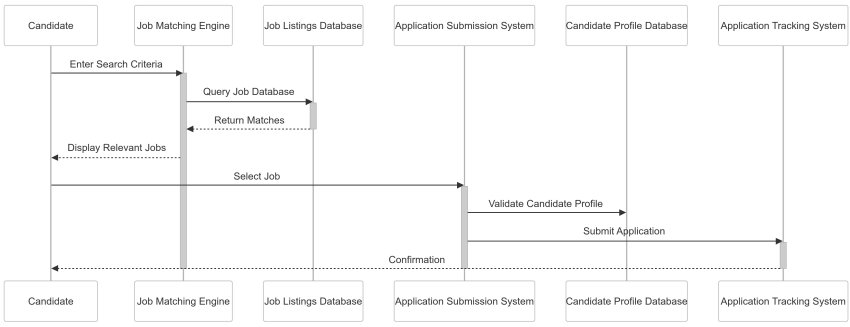


Figure A.4: SSD 2

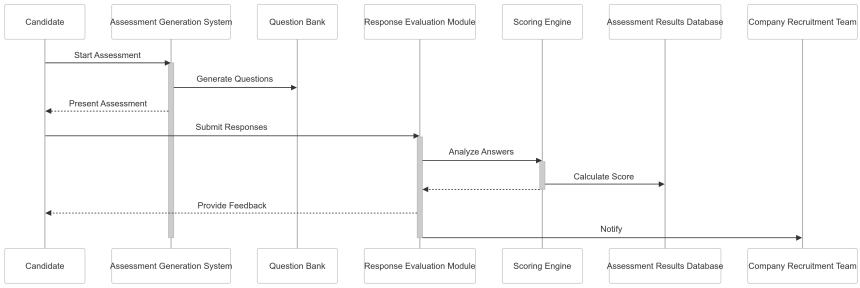


Figure A.5: SSD 3

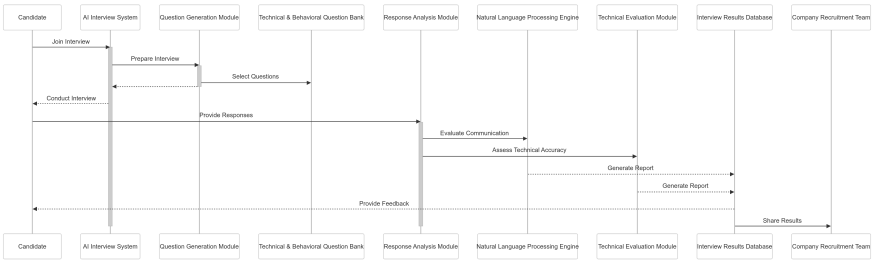


Figure A.6: SSD 4

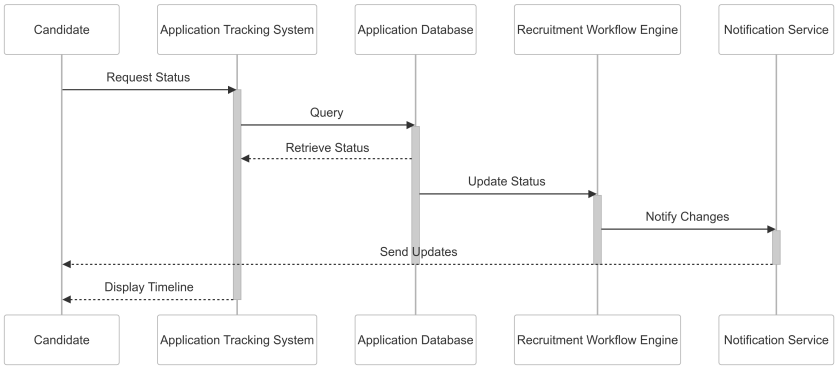


Figure A.7: SSD 5

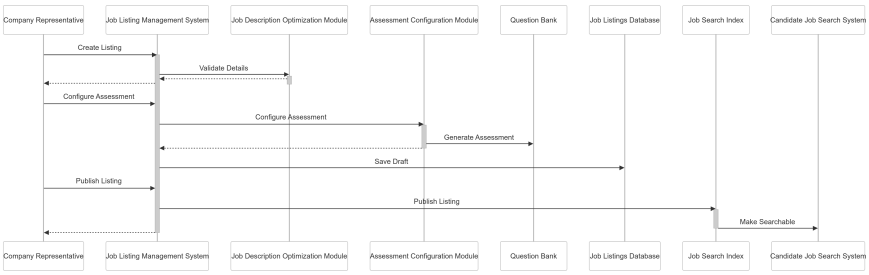


Figure A.8: SSD 6

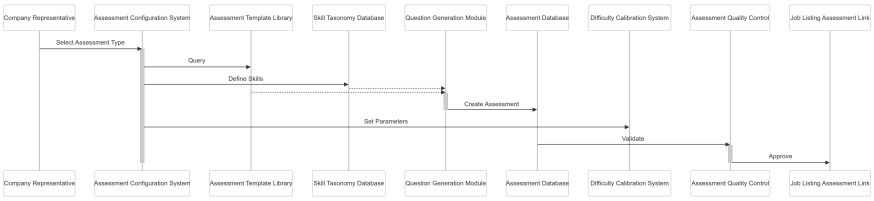


Figure A.9: SSD 7

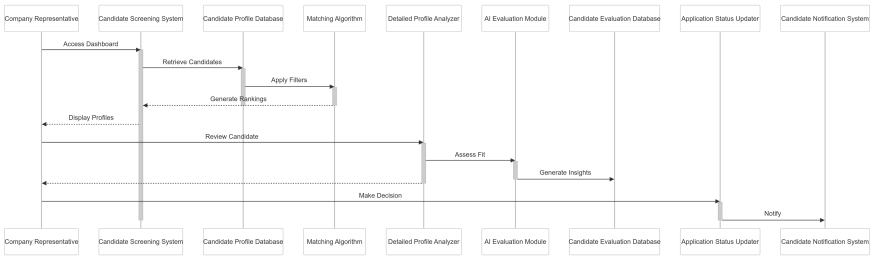


Figure A.10: SSD 8

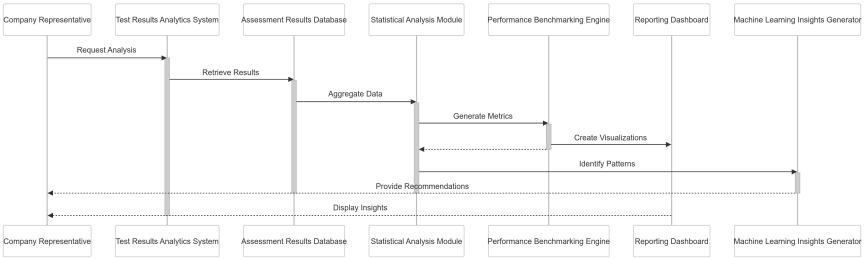


Figure A.11: SSD 9

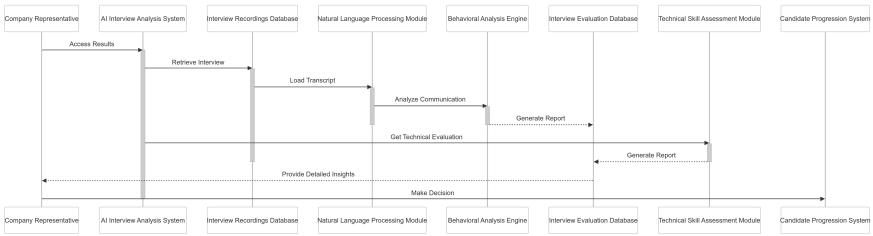


Figure A.12: SSD 10

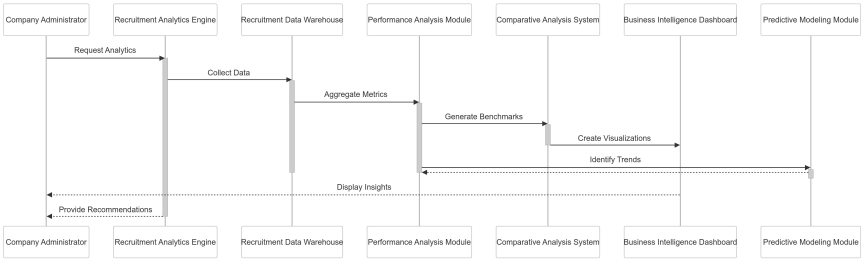


Figure A.13: SSD 11

A.4 Appendix D

A.4.1 Class Diagram

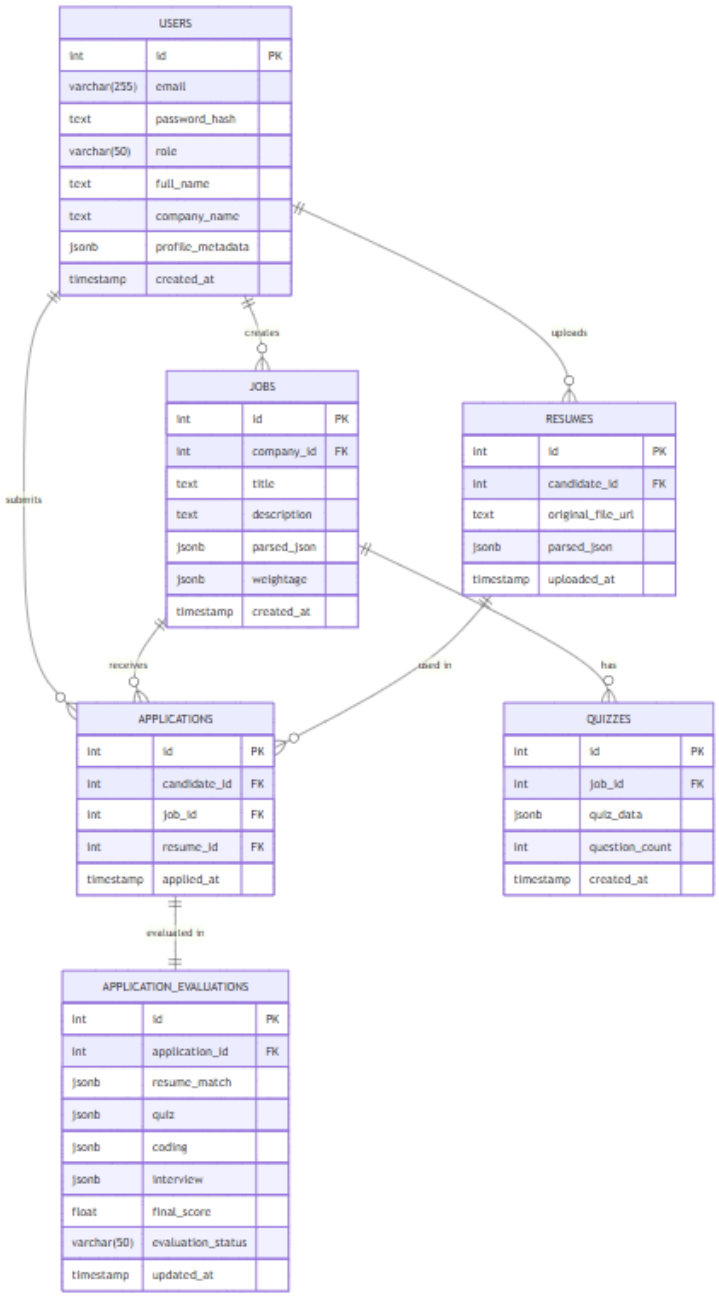


Figure A.14: Domain Model Diagram

A.4.2 Sequence Diagram

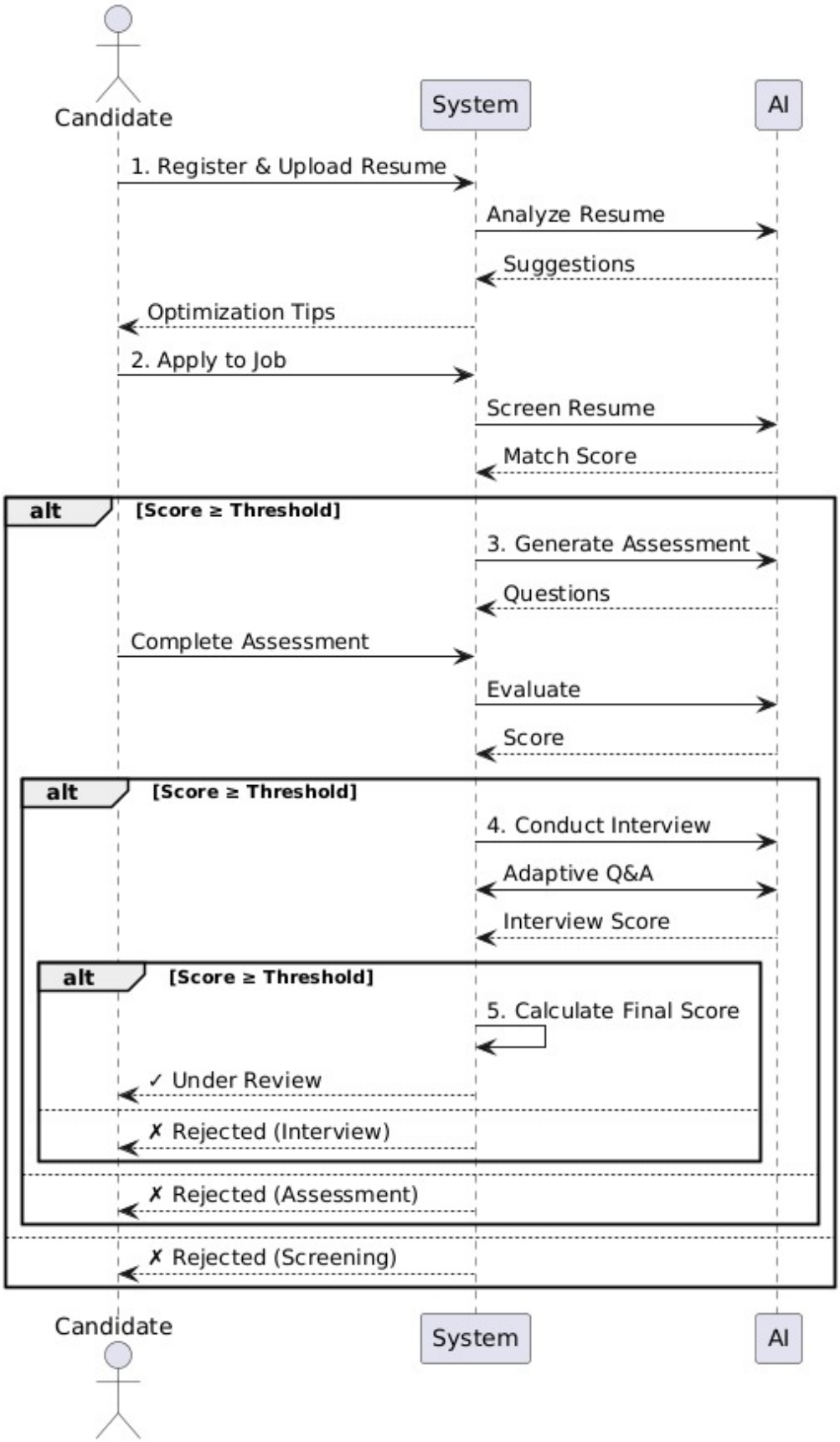


Figure A.15: State Diagram - Application Status Transitions